

at \$260,000 for a typical couple aged sixty-five. But there is large skew, and a 5% chance the couple will spend more than \$570,000. Many individuals also lack insurance—according to the Census Bureau, 21% of Americans earning between \$25,000 and \$49,000 per year do not have health insurance. Close to 20% of Americans between ages twenty-five and thirty-four do not have health insurance.³¹

Individuals can meet out-of-pocket health-care shocks and insure against the loss of income produced by poor health by holding safe assets. Health shocks are a form of *background risk*, which is the term given by economists to any risk on which individuals can't buy insurance.³² Background risk raises *precautionary savings* and the demand for safe assets.³³ Empirical evidence is consistent with this: Rosen and Wu (2004) show that people in poorer health hold fewer stocks and other risky assets. Life-cycle models with health-care risk, like De Nardi, French, and Jones (2010) and Pang and Warshawsky (2010), demonstrate that it is optimal to increase holdings of risk-free assets as retirement approaches. In retirement, individuals should optimally continue to save to meet out-of-pocket health-care expenses, whereas standard life-cycle models predict that individuals should dis-save.

Changing Utility

It's well established that people tend to become more risk averse as they get older.³⁴ (Perhaps this explains why your father used to attend anti-establishment rallies when young and then later switched political affiliations.) By itself, this would favor the 100 minus your age rule because, as the investor ages, he becomes more risk averse and wishes to hold fewer equities.

Endogenous Labor Income

Investment in labor income is a choice variable; that is, labor income is (partly) endogenous. Earning an MBA, for example, improves one's future earnings, but you trade off the cost of being out of the work force and paying tuition in exchange for the benefit of higher pay when you graduate. Labor income can also be increased by working longer hours or switching occupations. Shocks to wealth can be partially offset by working harder.³⁵ A young person has great flexibility, but that declines with age.

³¹ As reported by Rampini and Viswanathan (2013).

³² See Kimball (1990).

³³ See Hubbard, Skinner, and Zeldes (1995) and Palumbo (1999). Yogo (2011) shows that spending on medical care can reduce some health-care background risk.

³⁴ A recent review is Paulsen et al. (2012).

³⁵ When there is a stock market crash, the investor could consume less, work harder, or both. This labor margin makes a person effectively less risk averse, as Swanson (2012) notes. Other authors developing life-cycle models with endogenous labor income choices include Farhi and Panageas (2007) and Gomes, Kotlikoff, and Viceira (2008).